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Data Structure

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1. Circular Queue [Size = 6]

⇒ Enqueue(10), Enqueue(20), Enqueue(30), Dequeue(), Enqueue(40), Enqueue(50), Enqueue(60), Enqueue(70).

Step	Operation	Queue(0-5)	Front
Initial	-	[-,-,-,-,-,-]	-1
1.	Enqueue(10)	[10,-,-,-,-,-]	0
2.	Enqueue (20)	[10,20,-,-,-,-]	0
3.	Enqueue (30)	[10,20,30,-,-,-]	0
4.	Dequeue ()	[-,20,30,-,-,-]	1
5.	Enqueue (40)	[-, 20,30,40,-,-]	1
6.	Enqueue (50)	[-,20,30,40,50,-]	1
7.	Enqueue (60)	[-,20,30,40,50,60]	1
8.	Enqueue (70)	[70,20,30,40,50,60]	1

Final Queue:

20 → 30 → 40 → 50 → 60 → 70

Front Pointer = 1

Rear Pointer = 0

Explanation:

Initially, the queue is empty, The first Three enqueue operations insert 10, 20, 30 into the queue. When Dequeue() is performed, the first element (10) is removed, and the front pointer moves from index 0 to index 1.

2. Router Processes Packets Using a Queue:

Given:
* Packet Arrival order : P1, P2, P3, P4, P5.
* Buffer Size : 4 packets.

The Packets are stored as follows:

* P_1 - Stored
* P_2 - Stored
* P_3 - Stored
* P_4 - Stored
* P_5 - Cannot be stored because the buffer is full

(a) Identify Which Packet is Dropped:

The P5 packet is dropped:
Since the router's buffer has a maximum capacity of only four packets, it packets, it becomes full after storing P1, P2, P3, P4.

(b) Explain why the packet is discarded:

The packet is discarded because the router experience buffer overflow. A queue with a fixed buffer size cannot accept more packets once it becomes full. Since the queue follows the FIFO priciple, the existing packets are waiting to be removed until they are transmitted.

(c) Suggest a better queue management strategy to reduce packet loss.

A better queue management strategy is random early detection (RED) or Priority Queueing.

* Random Early Dection (RED):

RED monitors the buffuer ocupancy and begins dropping or marking packet before the buffer becomes completly full.

* Priority Queueing:

Packets are assigned different priority level based on their importance. This ensures better Quality of serive (QoS) & mimimizes the loss of critical packets.